Lab Quiz 1

CSNC2411 - Computer Communications and Networks

Total Marks: 10

Time Limit: 30 Minutes

Q1

Start Wireshark and simultaneously:

- open youtube for 10 seconds
- open google.com
- run:

```
tracert ucp.edu.pk
```

Now answer:

a) Which protocol generated the highest number of packets in YOUR capture? Explain WHY. **(1 mark)**

b) Find one TCP packet and one UDP packet.

For BOTH packets provide:

Field	TCP	UDP
Source Port	?	?
Destination Port	?	?
Header Size	?	?
Reliability Mechanism Present?	?	?

(2 marks)

c) In YOUR capture, identify:

- one application using UDP
- one application using TCP

Justify using packet evidence. **(1 mark)**

Q2

A student says:

"UDP is faster than TCP because UDP ignores acknowledgements."

Using YOUR Wireshark observations:

1. Explain why this statement is partially correct.
2. Explain one situation where UDP would FAIL badly.
3. Explain one situation where TCP would perform poorly compared to UDP.

(3 marks)

Q3

You observe the following in a TCP stream:

- SYN = 1
- ACK = 0
- Sequence Number = 4500

Questions:

1. Is this the first, second, or third step of TCP connection establishment?
2. What acknowledgement number should the receiver send back?
3. Why is ACK = 0 here?
4. What would happen if this packet gets lost?

(3 marks)

Instructions

- Attempt the quiz individually.
- Perform LIVE packet capturing using Wireshark during the quiz.
- Answers must be based on your own captured traffic and observations.
- Attach screenshots wherever required.
- Do not use previously saved `.pcap` files.
- Any copied answers/screenshots will be considered plagiarism.
- Submit your answers in PDF or DOC format on Google Classroom before the deadline.
- Late submissions will not be accepted.